

How to Protect ADS-B: Confidentiality Framework and Efficient Realization Based on Staged Identity-Based Encryption

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Abstract—Automatic Dependent Surveillance-Broadcast (ADS-B) is one of the key technologies for future “e-Enabled” aircrafts. ADS-B uses avionics in the e-Enabled aircrafts to broadcast essential flight data such as call sign, altitude, heading, and other extra positioning information. On the one hand, ADS-B brings significant benefits to the aviation industry, but, on the other hand, it could pose security concerns as channels between ground controllers and aircrafts for the ADS-B communication are not secured, and ADS-B messages could be captured by random individuals who own ADS-B receivers. In certain situations, ADS-B messages contain sensitive information, particularly when communications occur among mission-critical civil airplanes. These messages need to be protected from any interruption and eavesdropping. The challenge here is to construct an encryption scheme that is fast enough for very frequent encryption and that is flexible enough for effective key management. In this paper, we propose a Staged Identity-Based Encryption (SIBE) scheme, which modifies Boneh and Franklin’s original IBE scheme to address those challenges, that is, to construct an efficient and functional encryption scheme for ADS-B system. Based on the proposed SIBE scheme, we provide a confidentiality framework for future e-Enabled aircraft with ADS-B capability.

Index Terms—ADS-B, security, confidentiality, identity-based Encryption.

I. INTRODUCTION

A. e-Enabled Aircraft and ADS-B

THE purpose of future “e-Enabled” aircraft system is to enhance the efficiency of air transportation for both civil

and military aircraft [2], [3]. It is anticipated that all the current aircraft management systems will be upgraded to e-Enabled ones in the near future.

Automatic Dependent Surveillance-Broadcast (ADS-B) [4] is one of the essential components of the e-Enabled aircraft. It provides Air Traffic Controllers (ATCs) with up-to-date traffic information including altitude, speed and location of a concerned aircraft. This information is shared among different entities such as ground controllers, other aircraft and satellites. The information could also include real-time weather updates and temporary airspace borders. ADS-B can also be described as an Asynchronous Transfer Mode (ATM) technology that periodically broadcasts airplane status using a global navigation satellite system. ADS-B would improve safety by enhancing situational awareness of aircraft and boost flight efficiency by optimizing flight routes.

As discussed in [5], ADS-B has many advantages over the traditional radar-based technologies: First, ADS-B provides more accurate data than ground radar readings because the ADS-B messages are updated more frequently, broadcasting coordinates of airplane position approximately every 0.5 seconds. This will provide pilots with better and clearer directional information for takeoff, landing, and cruise. Second, in terms of cost, ADS-B needs less infrastructure as the use of the Global Positioning System (GPS) satellites, which cover a wider range of earth surface, allows the airborne objects to be monitored with fewer ground stations, and any entity that owns transmission receivers can capture and receive ADS-B messages.

Note that online applications could be installed in any smart devices or PCs to allow users to monitor the movement of the commercial aircraft through mashing up ADS-B data, as shown in Fig. 1.

Meanwhile, security of ADS-B system has been receiving a great deal of attention recently. For example, there have been discussions on various cyber-physical threats [7]–[11]. Also, authentication issues regarding the operations of ADS-B have been addressed [12]–[14]. In this paper, we focus on addressing confidentiality of ADS-B data. We notice that although ADS-B data are public in nature, providing ADS-B with confidentiality becomes critical for the environment where secrecy of ADS-B messages is demanded, such as military and diplomatic flights. However, we realized that there are challenges when confidentiality for ADS-B is provided. The first challenge is key management. Note that key management, which involves sharing keys, issuing and revoking (digital) certificates of public keys,

Manuscript received September 7, 2015; revised January 28, 2016, March 21, 2016, May 9, 2016, and June 18, 2016; accepted June 21, 2016. Date of publication August 5, 2016; date of current version February 24, 2017. This paper is an extended version of the work originally presented at IEEE INFOCOM Workshop 2015, titled “How to Protect ADS-B: Confidentiality Framework for Future Air Traffic Communication” [1]. This version includes complete security models for both ADS-B OUT and IN modes, and also detailed security and performance analyses. The Associate Editor for this paper was L. Li.

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Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TITS.2016.2586301

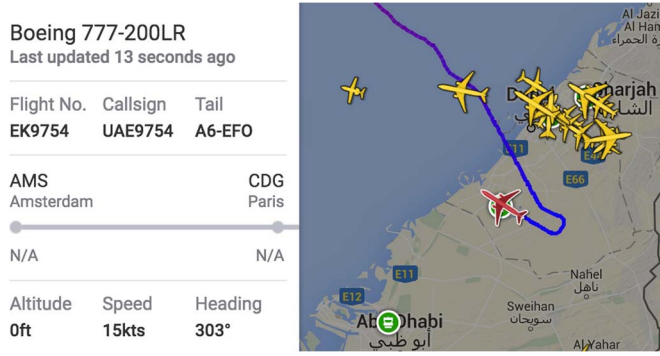


Fig. 1. Snapshot of a popular ADS-B online application, i.e., “Plane Finder” [6].

incurs a great amount of time and cost. The second challenge is how to realize highly efficient confidentiality service, which is of prime importance in ADS-B system where real-time communications are demanded.

B. Our Contributions and Organization of This Paper

The contributions of this paper are twofold:

- First, we propose a modified version of Boneh and Franklin’s IBE [15], which we call a “*Staged Identity-Based Encryption (SIBE)*” scheme, to address the aforementioned challenges of providing ADS-B with confidentiality. In addition to not requiring complex certificate management as it is identity-based, our scheme provides high level of efficiency by dividing encryption process into “key encryption” and “data encryption” and making key encryption perform only when a new “stage” is started.
- Second, we provide a confidentiality framework for ADS-B based on our SIBE scheme. Our framework will cover two modes of ADS-B operations, ADS-B OUT and ADS-B IN, which will be explained in detail in Section II.

The rest of the paper is organized as follows. In Section II, we provide detailed survey on ADS-B including related security issues. In Section III, we describe our SIBE scheme and confidentiality framework for ADS-B. Security and performance analyses of our framework are provided in Section IV, which is followed by concluding remarks in Section V.

II. MORE ON ADS-B AND ITS SECURITY

A. ADS-B Operations

ADS-B will be mandatory for the majority of aircraft in the U.S. to implement ADS-B from the beginning of 2020 [16]. ADS-B has already been enforced in countries like Australia, Canada, China, Sweden and UAE, some of which implemented ADS-B in their Air Traffic Management system.

ADS-B is transmitted at 1090 MHz or 978 MHz, mostly functioning on a Mode S transponder which is supplied with Aircraft Flight Management System (FMS). ADS-B signals sent from one aircraft to some ground controllers will be

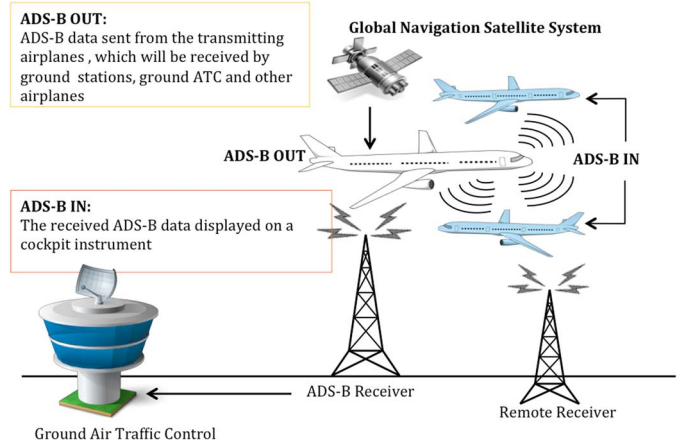


Fig. 2. Overview of the ADS-B.

rebroadcast to other aircraft in the area, and the same signal will be processed and sent to Air Traffic Control to be published online.

There are two services provided by ADS-B. The first service is Traffic Information Services-Broadcast (TIS-B), which is to provide air traffic information, and the second one is Flight Information Services-Broadcast (FIS-B), which is to mainly provide weather information. Due to the large bandwidth requirement of these signals, it uses the 978 MHz signal. The ADS-B signals can be displayed in aircraft Cockpit Multi-Function Display (CMFD) to be viewed by pilots to enhance situational awareness and improve the safety of the aircraft.

Fig. 2 illustrates an overview of ADS-B. As shown in the figure, many entities are considered in the system, and it is not limited to what is presented in the figure due to the scalability of the network. In the figure, “ADS-B OUT” indicates a communication mode where an aircraft equipped with ADS-B transponders send ADS-B messages to ground controllers and/or other aircraft, while “ADS-B IN” indicates a communication mode where an aircraft receives ADS-B messages from ground controllers and/or other aircraft.

In [2], [12] and [17], the authors present a systematic exposition of the different technologies and services that are related to ADS-B. In [8] and [18], the authors discuss the minimum technological standards and schemes needed in ADS-B to achieve a high level of performance.

B. ADS-B Message Structure

ADS-B messages are capable of providing ATC with highly efficient surveillance information on aircraft locations which are beyond the radar coverages. The maximum range of ADS-B is 250 NM (nautical miles) and the messages are sent every half second.

Two major standards for ADS-B data link are 1090 MHz Extended Squitter (1090ES) and Universal Access Transceiver (UAT). As illustrated in Fig. 3, the typical payload size of ADS-B message in UAT is 272 bits while 36 and 112 bits are allocated for synchronization information (SYNC) and forward error correction parity information (FEC PARITY) respectively [19]. Note that UAT is to support not only ADS-B but also

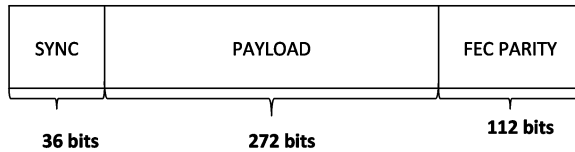


Fig. 3. ADS-B message format in UAT.

TIS-B and FIB-S. Note also that since UAT needs special hardware, it is used only in general aviation currently but it is expected to be deployed widely in the near future.

1090ES is based on the traditional Mode S system, widely used in commercial aircraft [20]. The payload size of ADS-B in 1090ES is 56 bits only. The ADS-B message format in 1090ES includes 5-bit DF (Downlink Format) to clearly indicate the type of message; 3-bit CA (read “Capability”) to indicate the capability of Mode S transponders or subtype; 24-bit ICAO address (or the tail number of the aircraft in question) and 24-bit PI (Parity Information).

C. The Current State of Security of ADS-B

Recently, a number of researchers have expressed their concerns about the security of ADS-B. In [20], the authors discussed that similar to any communication networks, ADS-B is vulnerable to many cyber attacks such as eavesdropping, jamming, message injection, modification and deletion. For example, aircraft reconnaissance [21] is a passive eavesdropping attack that could easily be performed over the open communication in ADS-B due to the unsecured channel. Furthermore, message modification can occur during the transmission phase via bit-flipping and overshadowing [22], [23].

The adversary can also be capable of external cyber physical exploits, such as active manipulation of data, node impersonation, radio-frequency jamming and internal (insider) exploits such as compromising cyber-elements. Credible examples of potential misuse by such an adversary in the future e-Enabled aircraft include malware infecting an aircraft system, exploitation of on-board wireless communications for unauthorized access to aircraft system interfaces, intentional denial of service of wireless interfaces to aircraft systems and misuse of off-board data links to access or disrupt aircraft system interfaces by introducing erroneous data mid-streamed between the aircraft and the ground station [5], [24]. Based on the vulnerability assessment, it would be important for aircraft to have security measures against cyber-physical vulnerabilities even if they are minimal. Most aviation standards do not yet deal with cyber-physical threats, but research community has begun to address this issue [7]–[10]. Especially, Costin and Francillon [11] demonstrated that passive attacks (eavesdropping) and active attacks (message jamming) on ADS-B are feasible to be conducted.

In this paper, we focus only on threats related to ADS-B data links and networks. Researchers are aware that ADS-B should be considered as an open broadcast system that could be threatened by many possible attacks. We note that Purton, Abbass and Alam identified ADS-B system vulnerabilities and threats by presenting a deep SWOT (strengths, weaknesses, opportunities,

and threats) analysis addressing network types, message spoofing and communication insufficiency [25]. According to this analysis, the most severe threat with high likelihood in ADS-B system is propagation induced errors, which occur due to weather and channel noise. Indeed, many previous works have been devoted to design and present authentication frameworks of the ADS-B communication, which are to prevent the error, misuse and modifications of the ADS-B messages. Especially, Valovage and Samuelson [12] considered a method utilizing Message Authentication Code (MAC). Ziliang, Weijun and Yang in [13] proposed a solution that uses Elliptic Curve Cryptography (ECC) as a public key encryption algorithm, providing a small 128-bit key, an X.509 certificate. More recently, Baek et al. proposed an authentication framework for ADS-B based on Online/Offline Identity-Based Signature (IBS) scheme [14].

In [20], Strohmeier, Lenders, and Martinovic provided the extensive survey on potential attacks on ADS-B and security measures that had been developed. They categorized the security solutions for ADS-B into two: non-cryptographic solutions based on the properties of the physical layer and cryptographic solutions based on public and secret (private) key cryptography. The authors raised concerns over the issue of difficulties related to key management and key distribution for the cryptographic solutions.

The authors of [2], [5], [10], [24], [26]–[28] focused on the threats on ADS-B and studied them from various aspects, such as ADS-B software/hardware, the key distribution algorithms, and discussed how security considerations and frameworks could be adopted in ADS-B [20], [29].

We note that no perfect solutions have yet been emerged for the security of ADS-B. Although [20] prefers non-cryptographic solution for the efficiency, we argue that the most secure solutions could be provided by cryptographic means.

III. PROPOSED CONFIDENTIALITY FRAMEWORK FOR ADS-B

As mentioned earlier, key management and efficiency are two important issues to address when designing confidentiality framework for ADS-B. To deal with the key management issue, we adopt the concept of Identity-Based Encryption (IBE) [15], [30], [31]. The rationale for using IBE is that since IBE uses receiving parties’ identities as public keys for encryption, it will greatly simplify the key management in public key encryption compared with traditional PKI-based key management which mandates digital certificates be used for authenticating every public key. However, this does not come free as many IBE schemes use pairings [15], which incur more computational cost compared with normal public key encryption. Since efficiency is also another important issue to be considered for ADS-B security, we cannot adopt IBE schemes as they are. Rather, we could take the approach of hybrid IBE [32], where data is encrypted by a symmetric encryption scheme like a block cipher under a chosen random key while the key is encrypted by IBE. Note that hybrid IBE is simply an extension of hybrid (public-key) encryption [33], [34]. Nevertheless, the hybrid IBE solution still may not be efficient enough for ADS-B applications. Hence, we further modify the hybrid IBE in such

a way that receiving parties remember the part of ciphertexts that encrypt the “staged” symmetric keys, so the key encryption which needs heavy pairing operation can be performed only at the beginning of each stage. Note that this is different from key exchange protocols which involve two-way communications and authentication between two entities. The details will follow in the subsequent subsection.

We remark that for realizing lightweight security services for ADS-B, one may consider to use a security protocol designed for wireless sensor network such as Perrig *et al.*’s Security Protocols for Sensor Networks (SPINS) [35]. However, the main difference between SPINS and ours is that we employ asymmetric cryptography, which does not require two communicating parties to share the same keys at the setup stage. The rationale behind this is that in the ADS-B setting, the distance between entities (ground controllers and aircraft) is usually very big, it may not be possible to share symmetric keys. Moreover, unlike SPINS, our solution does not need to divide time intervals and broadcast symmetric keys for each of the intervals, which makes key management quite complicated.

A. Overview of Staged Identity-Based Encryption

In this section, we describe our *Staged Identity-Based Encryption (SIBE)* scheme. As discussed earlier, the purpose of our scheme is to minimize heavy IBE operations by effectively utilizing the symmetric encryption scheme to achieve high overall efficiency. The main idea of the SIBE scheme is as follows. We assume that all of the ground controllers are identified by unique identifiers and have obtained private keys associated with them from the corresponding authority acting as a Private Key Generator (PKG). (Note that the International Civil Aviation Authority (ICAO) can have a role of the PKG to distribute private keys for the ground controllers.) When an aircraft enters an airspace managed by one ground controller, the aircraft encrypts a securely-generated random symmetric key with the identity of the ground controller. It then encrypts subsequent messages using a secure symmetric encryption scheme under the random key it had selected before. Upon receiving a ciphertext that encrypts the random symmetric key, which we call “key ciphertext,” the ground controller decrypts it using the private key associated with its identity and decrypt subsequent ciphertexts, which we call “data ciphertexts,” from the aircraft.

Having the list of identities of the ground controllers on its flight path, the aircraft will use each identity to encrypt a random symmetric key of its choice and the subsequent ADS-B messages will be encrypted by the symmetric key. When the aircraft communicates with a *new* ground controller on its flight path, a new symmetric key will be chosen and subsequent ADS-B messages for that ground controller will be encrypted by the new key. Note that the ground controller, which is in possession of the private key associated with its identity will be able to decrypt the encrypted key sent by the aircraft.

We note that combining symmetric encryption and IBE, i.e., constructing hybrid IBE is not new. Boneh and Franklin already discussed the possibility of using this technique in their original IBE scheme [15]. Since then, this technique has been employed

in numerous cryptographic schemes in the identity-based setting including Sheedy’s privacy-enhanced protocol [36] and Kiltz’ various IBE schemes [37]. Our SIBE scheme modifies Boneh and Franklin’s hybrid IBE scheme in such a way that key encryption and data encryption are performed in *different sessions* as described above to further improve efficiency. We found that this modification is suitable to deal with ADS-B communications where high efficiency is required.

B. Description of the Proposed SIBE Scheme Based on Boneh and Franklin’s IBE

We assume that our scheme is to be implemented on near-future ADS-B system like DO-260B [38]. To provide sound security level, the representation size of a group (the length of a group element in bits) should be 512 bits as recommended in [32]. We assume that cryptographic hash functions such as SHA1 or SHA2 is employed, and a secure block cipher with 128-bit key such as AES is used. We assume that this information about hash and encryption algorithms is pre-shared by involved parties. Alternatively, the information can be inserted in the payload and transmitted as plaintext before a first encryption is performed. Also, we assume that the PKG is assumed to be fully trusted while receiving parties (ground controller in ADS-B OUT mode and aircraft in ADS-B IN mode) are semi-trusted. This means that receiving parties may be curious about ADS-B messages which are not intended for them but they only listen to communications, obtain their private keys from the PKG and perform decryption correctly. (That is, they do not involve in active attacks like sabotaging communications.)

We now give the details of the proposed SIBE scheme.

- **Setup:** The PKG generates $\mathbb{G}_1$ and $\mathbb{G}_2$ of prime order q and an admissible pairing $e : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_2$, where g is a generator of $\mathbb{G}_1$. It then picks $s \in \mathbb{Z}_q$ at random and computes $y = g^s$. The PKG also selects a symmetric encryption scheme (such as AES) (E, D) , where E and D denote encryption and decryption procedures respectively. The PKG also selects two hash functions $H_1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$ and $H_2 : \mathbb{G}_2 \rightarrow \{0, 1\}^n$ for some positive integer n , which is the size of the key. (That is, $\{0, 1\}^n$ is key space.) The PKG sets secret master key $mk = s$ and a set of public parameters $params = (\mathbb{G}_1, \mathbb{G}_2, e, g, y, (E, D), H_1, H_2)$. The PKG distributes $params$ to all of the parties involved.
- **Extract:** Upon receiving an identity ID , the PKG computes $H_1(ID)^s \in \mathbb{G}_1$ and returns $K_{ID} = H_1(ID)^s$ as a private key associated with ID .
- **KEYEncrypt:** The sender performs the following.
 - Pick a symmetric key k uniformly at random from the key space $\{0, 1\}^n$.
 - Pick $r \in \mathbb{Z}_q$ uniformly at random.
 - Compute $u = g^r$ and $v = H_2(e(H_1(ID), y)^r) \oplus k$.
 - Return $C_k = (u, v)$ as “key ciphertext.”
- **DATAEncrypt:** Having selected the random symmetric key k , the sender performs the following.
 - Compute $C_M = E_k(M)$.
 - Return C_M as “data ciphertext.”

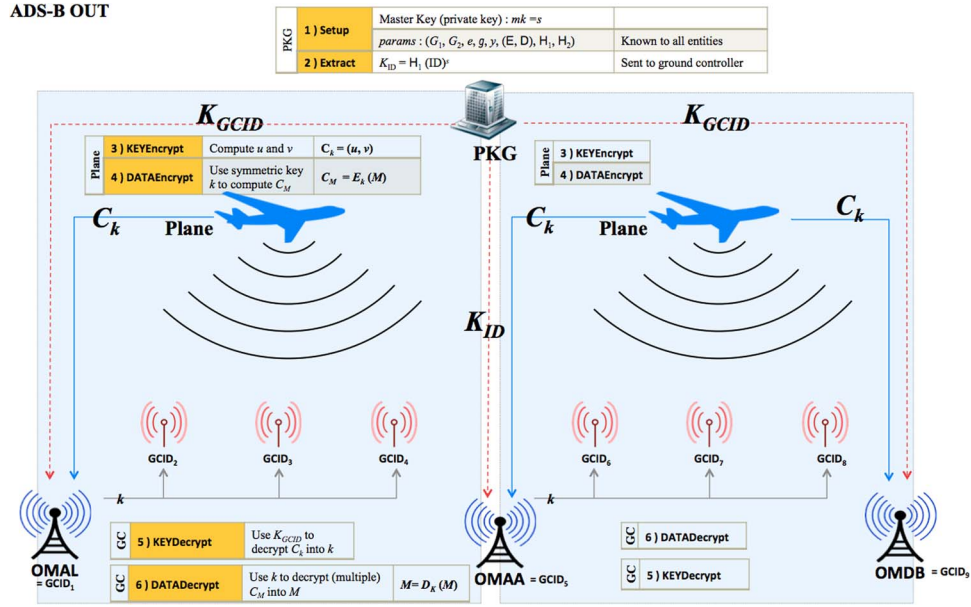


Fig. 4. Use case scenario of our confidentiality framework for the ADS-B OUT mode based on SIBE.

- **KEYDecrypt:** Upon receiving key ciphertext $C_k = (u, v)$, the receiver performs the following using K_{ID} , the private key associated with ID.
 - Compute $k = v \oplus H_2(e(K_{ID}, u))$.
 - Return k .
- **DATADecrypt:** Upon receiving data ciphertext C_M , the receiver performs the following using k , the decrypted symmetric key.
 - Compute $M = D_k(C_M)$.
 - Return M .

Note that the pairing function e is assumed to be symmetric pairing and satisfies the following properties [32]: 1) Non-degeneracy: $e(g, g) \neq 1 \in \mathbb{G}_2$; 2) Bilinearity: For all $a, b \in \mathbb{Z}_p$, $e(g^a, g^b) = e(g, g)^{ab}$.

C. Use Case Scenario

In this subsection, we present a use case scenario of our confidentiality framework based on SIBE in the ADS-B OUT Mode.

1) Use Case Scenario for ADS-B OUT Mode: (See Fig. 4)

- The air transportation authority or a similar government body acts as a Private Key Generator (PKG). By running the Setup algorithm, the authority first produces the master key mk_A and the public parameters denoted by $params_A$. It is assumed that the validity of $params_A$ has been checked properly (possibly by checking the certificate signed by another trusted third party). Next, $params_A$ is distributed to e-Enabled aircraft, ground controllers and any other interested parties. The lifetime of $params_A$ is usually long, e.g., one year.
- Each ground controller will be identified by a GCID (Ground Controller ID), which is of the following format: $[GCID = AN||CT||CN||AUX]$ where AN, CT, CN

and AUX denote a nearby Airport Name (AN) identified by ICAO code, Controller Type (CT) such as air traffic control tower and Air Route Traffic Control Center (ARTCC), Controller Number (CN) and other Auxiliary information (AUX), respectively. In Fig. 4, for instance, “OMAL” denotes the ICAO code of Al Ain Airport, “OMAA” for Abu Dhabi Airport and “OMDB” for Dubai Airport.

- To encrypt its data in ADS-B OUT mode, an aircraft first obtains a list of identities (GCID’s) of ground controllers. Note that this is possible because almost all commercial flights are governed by IFR (Instrument Flight Rule), flight paths are predetermined and preloaded in the FMS (Flight Management System) in the aircraft before each flight. We assume that the aircraft retains a list of identities of ground controllers in the flight path, denoted $GCID_1, GCID_2, \dots, GCID_n$. We assume that $GCID_1$ and $GCID_n$ are departing and arriving airports respectively. (Therefore, $GCID_1 = OMAL$ and $GCID_8 = OMDB$ in Fig. 4.) The aircraft then will encrypt its ADS-B OUT data as follows:

- 1) For the first time the aircraft communicates with a ground controller identified by $GCID_i$ for $1 \leq i \leq n$, it runs the KEYEncrypt algorithm to encrypt a random symmetric key denoted by k_i . It transmits the key ciphertext denoted by (u_i, v_i) to the ground controller.
- 2) To encrypt the ADS-B OUT data, the aircraft runs the DATAEncrypt algorithm to encrypt subsequent ADS-B OUT data until it encounters the next ground controller, identified by $GCID_{i+1}$.
- 3) Steps 1 and 2 are repeated until the aircraft lands, communicating with the final ground controller identified by $GCID_n$ (normally a control tower at the destination airport).

ADS-B IN

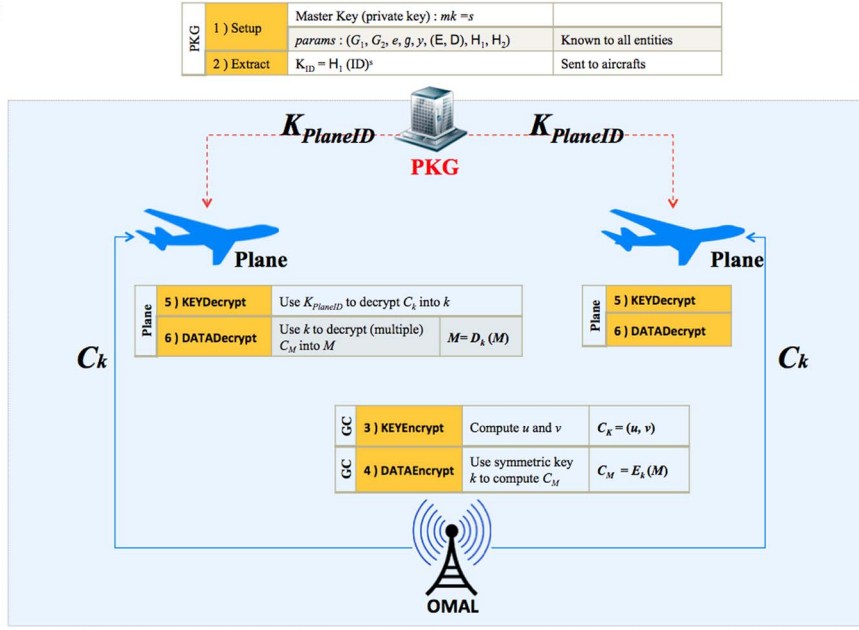


Fig. 5. Use case scenario of our confidentiality framework for the ADS-B IN mode based on SIBE.

- 4) A ground controller identified by $GCID_i$, which receives key ciphertext (u_i, v_i) and data ciphertext C_M , runs the **KEYDecrypt** algorithm to recover the encrypted symmetric key k_i and runs the **DATADecrypt** algorithm to decrypt C_M into M using k_i .

Note that in contrast to the regular public keys that come with digital certificates, $GCID$'s are merely strings, which are much easier to handle and to preload to FMS. Note also that the IBE will provide *implicit* certification of $GCID$ by having the air transportation authority (acting as the PKG) generate the private key associated with $GCID$'s.

2) Use Case Scenario for ADS-B IN Mode: (See Fig. 5)

- The dispatch of an airline company, for instance, acts as a PKG. It is assumed that the dispatch has produced a master key msk_D and the public parameters $params_D$ by running the **Setup** algorithm. It is also assumed that $params_D$ is trusted and has been distributed to e-Enabled aircraft, ground controllers and any other parties involved.
- Each aircraft will be identified by $PlaneID$, which is of the following format: $[PlaneID = AR||AD||FN||AUX]$ where AR, AD, FN and AUX denote Aircraft Registration (AR), Airline Designator (AD) such as the 24-bit ICAO address, Flight Number (FN) and Auxiliary information (AUX) respectively. We assume that AUX may contain any additional information needed to uniquely identify an aircraft, such as the date and duration of the flight.

- 1) Ground controllers and other aircraft will use $PlaneID$ to encrypt a random symmetric key by running the **KEYEncrypt** algorithm. Then they encrypt subsequent ADS-B IN data by running the **DATAEncrypt** under the random key it generated before. They send key ciphertext (u, v) and data ci-

phertext C_M to the aircraft identified by $PlaneID$. Note that the generation of key ciphertext (which is computationally more expensive) is less frequent than the generation of data ciphertext (which is computationally much less expensive).

- 2) Upon receiving the key ciphertext (u, v) and data ciphertext C_M from ground controllers or other aircraft, an aircraft identified by $PlaneID$ runs the **KEYDecrypt** algorithm to recover the encrypted symmetric key k and runs the **DATADecrypt** algorithm to decrypt C_M into M using k .

IV. SECURITY AND PERFORMANCE ANALYSIS

A. Security Analysis

In this section, we analyze the security of the proposed SIBE scheme. First, we define the notion of indistinguishability of the SIBE scheme against chosen plaintext attack, which we call "IND-SIBE-CPA."

Definition 1: Let $SIBE = (\text{Setup}, \text{Extract}, \text{KEYEncrypt}, \text{DATAEncrypt}, \text{KEYDecrypt}, \text{DATADecrypt})$ be an SIBE scheme. Let $\mathcal{A}$ be an adversary that is a probabilistic polynomial-time (PPT) algorithm. Assume that $\mathcal{A}$ has access to $params$, which has been created by the **Setup** algorithm. Then $\mathcal{A}$ interacts with the "Challenger," performing the following steps:

- 1) If $\mathcal{A}$ queries identities, each of which is denoted by ID , the Challenger provides $\mathcal{A}$ with the corresponding private keys K_{ID} 's obtained by running the **Extract** algorithm.
- 2) $\mathcal{A}$ outputs two equal-length plaintexts (M_0, M_1) and a target identity ID^* .
- 3) The Challenger picks a random key k^* from the appropriate key space, chooses $\beta \in \{0, 1\}$ at random (i.e., with probability $1/2$), encrypts k^* under ID^* and encrypts

M_β using the symmetric encryption algorithm E under k^* . The resulting key ciphertext and data ciphertext pair $(C_{k^*}^*, C_{M_\beta}^*)$ is sent to $\mathcal{A}$.

- 4) $\mathcal{A}$ obtains private keys of identities ID 's of its choice from the Challenger with the restriction that $ID \neq ID^*$.
- 5) $\mathcal{A}$ outputs a guess $\beta' \in \{0, 1\}$ (on β).

We define $\text{Adv}_{\text{SIBE}, \mathcal{A}}^{\text{ind-sibe-cpa}}(\lambda) = |\Pr[\beta' = \beta] - (1/2)|$, where λ is a security parameter and the probability is taken over the random coins used by $\mathcal{A}$ and the random coins used by the Challenger. The scheme **SIBE** is IND-SIBE-CPA secure if, for all $\mathcal{A}$, there exists a negligible function $\text{negl}(\lambda)$ such that $\text{Adv}_{\text{SIBE}, \mathcal{A}}^{\text{ind-sibe-cpa}}(\lambda) \leq \text{negl}(\lambda)$.

Before presenting the main result regarding the security of the **SIBE** scheme, we review the following definitions.

Definition 2: Let $\text{SYM} = (\text{KG}, E, D)$ be a symmetric encryption scheme. Let $\mathcal{A}$ be a PPT adversary. Assume that a symmetric key k has been generated uniformly at random by the **KG** (key generation) algorithm. (Note that $\mathcal{A}$ does not have access to this key.) Then, $\mathcal{A}$ interacts with the ‘‘Challenger,’’ performing the following steps:

- 1) If $\mathcal{A}$ queries any messages, each of which is denoted by M , for encryption, the Challenger provides $\mathcal{A}$ with the corresponding ciphertext C obtained by running the encryption algorithm E with k .
- 2) $\mathcal{A}$ outputs two equal-length plaintexts (M_0, M_1) .
- 3) The Challenger picks one of (M_0, M_1) at random (with probability $1/2$) and encrypts M_β using the symmetric encryption algorithm E under k , where β is a random bit. The resulting ciphertext C^* is sent to $\mathcal{A}$.
- 4) $\mathcal{A}$ queries plaintexts of its choice for encryption.
- 5) $\mathcal{A}$ outputs a guess $\beta' \in \{0, 1\}$ (on β).

We define $\text{Adv}_{\text{SYM}, \mathcal{A}}^{\text{ind-cpa}}(\lambda) = |\Pr[\beta' = \beta] - (1/2)|$, where λ is a security parameter and the probability is taken over the random coins used by $\mathcal{A}$, as well as the random coins used by the Challenger. The scheme **SYM** is IND-CPA secure if for all $\mathcal{A}$, there exists a negligible function $\text{negl}(\lambda)$ such that $\text{Adv}_{\text{SYM}, \mathcal{A}}^{\text{ind-cpa}}(\lambda) \leq \text{negl}(\lambda)$.

We also need the following definition of the Bilinear Diffie-Hellman (BDH) problem.

Definition 3: Let $\mathcal{A}$ be a PPT adversary. Suppose that $\mathbb{G}_1$ and $\mathbb{G}_2$ are groups of prime order q and there exists an admissible pairing $e : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_2$, where g is a generator of $\mathbb{G}_1$. Assume that $\mathcal{A}$ is given as (g, g^a, g^b, g^c) , where a, b and c are chosen uniformly at random from $\mathbb{Z}_q$. The BDH problem is a computational problem that $\mathcal{A}$ finds $e(g^a, g^b)^c (= e(g, g)^{abc})$. The BDH problem is hard if there exists a negligible function $\text{negl}(\lambda)$ such that $\mathcal{A}$'s advantage $\text{Adv}_{\mathbb{G}_1, \mathbb{G}_2, \mathcal{A}}^{\text{bdh}}(\lambda) \leq \text{negl}(\lambda)$.

We now state the following theorem regarding the confidentiality of the proposed **SIBE** scheme.

Theorem IV.1: The proposed **SIBE** scheme is IND-ID-CPA secure in the random oracle model assuming that the BDH problem is hard and the underlying symmetric encryption scheme **SYM** is IND-CPA secure.

Proof: Let $\mathcal{B}$ be an adversary to solve the BDH problem. Let $\mathcal{A}$ be an adversary that attacks the confidentiality of the

proposed **SIBE** scheme. We show how $\mathcal{B}$ can solve the BDH problem using $\mathcal{A}$ as a subroutine.

Suppose that $\mathcal{B}$ is given (g, g^a, g^b, g^c) . First, $\mathcal{B}$ sets $y = g^b$ and gives y to $\mathcal{A}$ as a public parameter. $\mathcal{B}$ then picks an index t at random from $\{1, \dots, q_{H_1}\}$, where q_{H_1} denotes the (maximum) number of queries to the random oracle H_1 .

Whenever $\mathcal{A}$ submits ID as a query to the random oracle H_1 , $\mathcal{B}$ picks $\mu \in \mathbb{Z}_q$ at random and returns $H_1(ID) = g^\mu$. Note that $\mathcal{B}$ maintains a list of query-answers for the random oracle H_1 , each of its entry is denoted by $\langle ID, \mu, g^\mu \rangle$.

Whenever $\mathcal{A}$ submits ID as a private key extraction query, $\mathcal{B}$ searches an entry $\langle ID, \mu, g^\mu \rangle$ of the query-answer list for the random oracle H_1 and returns y^μ to $\mathcal{A}$ as an answer to the private key extraction query. (Note that y^μ is indeed a valid private key since $[y^\mu = g^{b\mu} = H_1(ID)^b]$).

Upon receiving (ID^*, M_0, M_1) , where ID^* is the challenge identity, $\mathcal{B}$ first checks whether ID^* is $\mathcal{A}$'s t -th query to the random oracle H_1 . If it is not, $\mathcal{B}$ simply tosses a random coin $\beta \in \{0, 1\}$, outputs β and halts. Otherwise, $\mathcal{B}$ sets $H_1(ID^*) = g^c$ and continues the simulation as follows. $\mathcal{B}$ picks $\beta \in \{0, 1\}$ at random; sets $u^* = g^a$; picks v^* and k^* uniformly at random from $\{0, 1\}^n$. $\mathcal{B}$ then sets $H_2(e(g^c, g^b)^a) = v^* \oplus k^*$ and returns $(u^*, v^*, E_{k^*}(M_\beta))$ as a challenge ciphertext. This is a perfect simulation of the challenge ciphertext as $u^* = g^a$ and $[v^* = k^* \oplus H_2(e(g^c, g^b)^a) = k^* \oplus H_2(e(H_1(ID^*), y)^a)]$.

$\mathcal{B}$ continues to simulate answers for the random oracle and private key extraction queries as before. When $\mathcal{A}$ outputs its guess β' on $\beta \in \{0, 1\}$, $\mathcal{B}$ halts. Additionally, at any time, the random oracle H_1 is queried by $e(g^a, g^b)^c$.

Let E_1 be an event in which H_1 is not queried by $e(g^a, g^b)^c$. Now, consider the event $\beta' = \beta$. Note that in the above simulation, as long as H_1 is not queried by $e(g^a, g^b)^c$, the symmetric key k (output of H_1) is randomly and independently distributed, the event $\beta' = \beta$ leads to an attack on the confidentiality of the symmetric encryption scheme **SYM**. In other words, one can construct an IND-CPA adversary $\mathcal{C}$ for the symmetric encryption **SYM**, which can simulate the environment of $\mathcal{A}$ perfectly assuming that E_1 does not happen. In other words, the simulation and the real attack for **SYM** is indistinguishable, so we have $|\Pr_{\text{sim}}[\beta' = \beta] - \Pr_{\text{real}}[\beta' = \beta]| \leq \Pr[E_1]$, where $\Pr_{\text{sim}}$ denotes the probability distribution in a simulated environment in which $\mathcal{A}$ resides, created by $\mathcal{C}$, and $\Pr_{\text{real}}$ denotes the probability distribution in the real attack for **SYM**. Since $\Pr_{\text{real}}[\beta' = \beta] = (1/2) + (1/2)\text{Adv}_{\text{SYM}, \mathcal{C}}^{\text{ind-cpa}}(\lambda)$ by definition, we have $\Pr_{\text{sim}}[\beta' = \beta] \leq (1/2) + (1/2)\text{Adv}_{\text{SYM}, \mathcal{C}}^{\text{ind-cpa}}(\lambda) + \Pr[E_1]$.

Now, let E_2 be an event in which the challenge identity ID^* is $\mathcal{A}$'s t -th query to the random oracle H_1 . If we split the event $\beta = \beta'$ in the simulation, we have $\Pr_{\text{sim}}[\beta' = \beta] = \Pr_{\text{sim}}[\beta' = \beta | E_2] \Pr_{\text{sim}}[E_2] + \Pr_{\text{sim}}[\beta' = \beta | \neg E_2] \Pr_{\text{sim}}[\neg E_2] = \Pr_{\text{sim}}[\beta' = \beta | E_2] \Pr_{\text{sim}}[E_2] + (1/2) \Pr_{\text{sim}}[\neg E_2] = (1/2) + \Pr_{\text{sim}}[E_2] (\Pr_{\text{sim}}[\beta' = \beta | E_2] - (1/2))$.

Note that $\Pr_{\text{sim}}[E_2] = (1/q_{H_1})$ and $\Pr_{\text{sim}}[\beta' = \beta | E_2] = \Pr[\beta' = \beta] = (1/2) + (1/2)\text{Adv}_{\text{SIBE}, \mathcal{A}}^{\text{ind-sibe-cpa}}(\lambda)$ by the way the challenge ciphertext is constructed in the simulation. (Note that if E_2 happens, the challenge ciphertext that is constructed in the simulation is indistinguishable from the one in the real

attack.) Now, since the event E_1 implies solving the BDH problem, we have $\Pr[E_1] = \text{Adv}_{\mathbb{G}_1, \mathbb{G}_2, \mathcal{B}}^{\text{bdh}}(\lambda)$. Then, combining all of these, we obtain the following bound:

$$\text{Adv}_{\text{SIBE}, \mathcal{A}}^{\text{ind-id-cpa}}(\lambda) \leq q_{H_1} \text{Adv}_{\mathbb{G}_1, \mathbb{G}_2, \mathcal{B}}^{\text{bdh}}(\lambda) + q_{H_1} \text{Adv}_{\text{SYM}, \mathcal{C}}^{\text{ind-cpa}}(\lambda).$$

This concludes the proof. $\square$

We remark that the IND-CPA requirement of the symmetric encryption scheme SYM implies that each encryption is randomized so that “indistinguishability” of ciphertexts is retained (even if the same message is encrypted twice, for instance). A number of ways to construct IND-CPA secure encryption are known, e.g., $E(k, m) = s \parallel (F(k, s) \oplus m)$, where F is a pseudorandom function, which can be constructed using a secure block cipher like AES, k is a shared key, and s is a random string (See Section 3.5.2 in [33]). Note that our SIBE implementation is based on this scheme.

We also remark that security against chosen ciphertext attack, i.e., IND-CCA [39], can also be considered, which would be based on the chosen ciphertext security for IBE presented in [15]. However, realizing chosen ciphertext security would need more overhead in terms of computation and communication, so we focused only on chosen plaintext security of SIBE for the sake of efficiency.

B. Performance Analysis

As discussed above, the efficiency gains of the proposed confidentiality framework stem from the fact that the heavy IBE computation is only performed at the beginning of each stage when the plane begins to communicate with a new ground controller. In order to demonstrate the performance, we implemented our scheme on a PC with a 1.8-GHz Intel Core i5 CPU and 4GB memory. We used the Java-based pairing computation package called “JPair” developed by Dong [40]. We used the supersingular curve with 80-bit security, so the representation size of the element in $\mathbb{G}_1$ is 512 bits as in Boneh and Franklin’s original IBE scheme [15]. We used SHA1 for the hash function H_2 , whose size is 160 bits. We assumed that the ADS-B message format follows the UAT specification, which is discussed in Section II-B.

In an experiment to measure performance, we ran the proposed SIBE scheme to simulate a situation in which a plane starts to communicate to a new ground controller and sends confidential messages 50 times. (We encrypt the same messages fifty times, but in reality the plane needs to encrypt 50 different message of the same size with have slightly different content.) We set the identity as $\text{ID} = \text{AUH_TOWER_0001_MAY2014}$ and encrypt the message $M = \text{a39125|s417|h324}$. (Note that units for altitude (a), speed (s) and heading (h) are feet, knots and degrees respectively.) Namely, in the experiment, it is assumed that the key encryption is performed once using the given identity ID and data encryption with respect to this identity is conducted fifty times subsequently. We then ran the regular hybrid IBE scheme to encrypt the message 50 times in the same situation using the same identity ID. We compared the running time of SIBE with that of the normal IBE.

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TABLE I
TOTAL RUNNING TIME OF IBE AND SIBE (OUR SCHEME) WHEN FIFTY
CONFIDENTIAL MESSAGES ARE TRANSMITTED

Scheme	IBE	SIBE (Our Scheme)
Running Time	91649.21 ms	10519.53 ms

We note that as discussed in [13], implementations of any cryptographic algorithm for ADS-B would need modification of data format of ADS-B so does our implementation. As the typical block size of payload in the UAT data link of ADS-B is 272 bits (as illustrated in Fig. 3). Since we reserve 16 bits for inserting timestamp to prevent replay attack as done in [13], the actual size of data payload is only 256 (= 272 – 16) bits. Meanwhile, the size of key ciphertext (u, v) is 672 bits. Hence, we need $3 (= \lceil (672/256) \rceil)$ ADS-B data blocks to transmit each key ciphertext, which will cause slight delay of ADS-B message transmission. However, the block size of data ciphertext is 128 bits when AES is used, the size of a message including a string for randomization is up to 256 bits, which can be packed into one data block of ADS-B in UAT.

In Table I, we summarize the total encryption time of IBE and SIBE in this experiment. Moreover, Fig. 6 illustrates the encryption time of IBE and SIBE for encrypting 50 messages. Note that SIBE performs much better than normal hybrid IBE due to the fact that the key encryption that involves pairing operation is conducted only when the stage is changed (that is, a plane starts to communicate with new ground controllers) From our implementation, it is estimated that it is nearly 9 times faster than the normal hybrid IBE. Based on this performance, it is highly feasible for our scheme to be used in a real-world situation: Note that it is estimated that the maximum time delay allowed for a ADS-B-Out message transmission (throughput) is $T_{\max} = 500$ ms (milliseconds).

Now, let t_{IBE} and t_{SIBE} be the average encryption times of the normal hybrid IBE and our SIBE schemes, respectively. Then, we have $t_{\text{IBE}} = 91649.21/50 = 1832.98$ ms and $t_{\text{SIBE}} = 10519.53/50 = 210.39$ ms from Table I. Therefore, we have $t_{\text{SIBE}} < T_{\max} < t_{\text{IBE}}$. Although the encryption time reduces room for transmission delay, it is well below the maximum time delay for ADS-B message transmission, so the proposed SIBE scheme seems feasible for ADS-B in practice. However, we remark that we do not differentiate the delays of transmission between plane-to-plane and plane-to-ground controller. Considering various factors of transmission for the SIBE scheme will be our future research.

C. Additional Note on Performance

We note that it would not be difficult to replace the key encryption algorithm with any other IBE, e.g., Boneh and Boyen’s IBE [41]. When Boneh and Boyen’s scheme is used, we can obtain more efficient key encryption. The main saving comes from the hash function H_1 , which does not need to map an input string to a point on elliptic curve unlike Boneh and Franklin’s IBE. Below, we describe this modification.

- **Setup:** The PKG generates $\mathbb{G}_1$ and $\mathbb{G}_2$ of prime order q and an admissible pairing $e : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_2$, where g is a generator of $\mathbb{G}_1$. It then picks $\alpha, \beta, \gamma \in \mathbb{Z}_q$

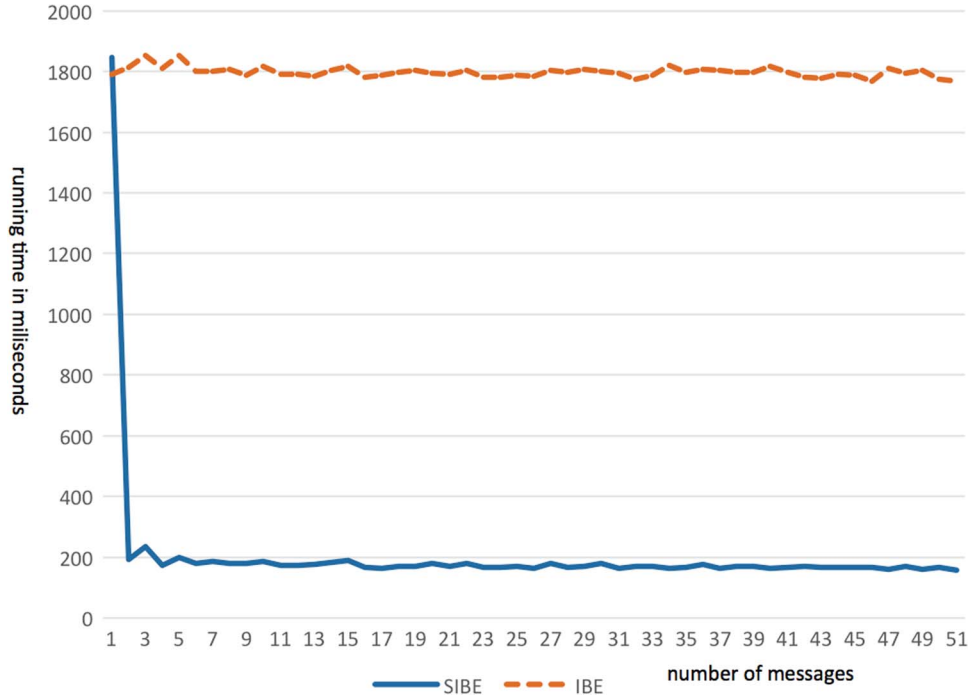


Fig. 6. Comparison of the running time of SIBE and (Hybrid) IBE.

at random and computes $y_1 = g^\alpha$, $y_2 = g^\beta$, $y_3 = g^\gamma$ and $\omega = e(g, g)^{\alpha\beta}$. The PKG also selects a symmetric encryption scheme (E, D) . The PKG also selects two hash functions $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ and $H_2 : \mathbb{G}_2 \rightarrow \{0, 1\}^n$ for some positive integer n , which is the size of the key. The PKG sets secret master key $mk = (\alpha, \beta, \gamma)$ and a set of public parameters $params = (\mathbb{G}_1, \mathbb{G}_2, e, g, y_1, y_2, y_3, \omega, (E, D), H_1, H_2)$. The PKG distributes $params$ to all of the parties involved.

- **Extract:** Upon receiving an identity ID , the PKG returns $K_{ID} = (g^{\alpha\beta + (\alpha H_1(ID) + \gamma)t}, g^t)$ for random $t \in \mathbb{Z}_q$ as a private key.
- **KEYEncrypt:** The sender performs the following.
 - Pick a symmetric key k uniformly at random from the key space $\{0, 1\}^n$.
 - Compute $u = g^r$, $v = y_3^r y_1^{H_1(ID)r}$ and $w = H_2(\omega^r) \oplus k$ for random $r \in \mathbb{Z}_q$, and return $C_k = (u, v, w)$ as “key ciphertext”.
- **KEYDecrypt:** Upon receiving key ciphertext $C_k = (u, v, w)$, the receiver performs the following using $K_{ID} = (d_1, d_2)$, the private key associated with ID .
 - Compute $k = w \oplus H_2(e(u, d_1)/e(v, d_2))$ and return k .

Note that $e(u, d_1)/e(v, d_2) = e(g^r, g^{\alpha\beta + (\alpha H_1(ID) + \gamma)t}) / e(y_3^r y_1^{H_1(ID)r}, g^t) = e(g^r, g^{\alpha\beta}) e(g^r, g^{(\alpha H_1(ID) + \gamma)t}) / e(g^{\gamma r} g^{\alpha H_1(ID)r}, g^t) = e(g^r, g^{\alpha\beta}) e(g^r, g^{(\alpha H_1(ID) + \gamma)t}) / e(g^{(\gamma + \alpha H_1(ID))r}, g^t) = e(g^r, g^{\alpha\beta}) = g(g^\alpha, g^\beta)^r = \omega^r$. Therefore, the KEYDecrypt algorithm is correct. Note also that since DATAEncrypt and DATADecrypt are the same as those described in Section III-B, they are omitted here.

However, care must be taken when interpreting the efficiency that various IBE schemes offer. That is, we may need to sacrifice security to achieve better performance, which is the

case in the above scheme: The security of Boneh and Boyen’s scheme against chosen plaintext attack is proven so called “selective-ID” model [41], which assumes that an adversary needs to commit to a particular identity ahead of the attack. This assumption is indeed stronger than the assumption the security of Boneh and Franklin’s original scheme is based on.

We remind readers that the purpose of this paper is *not to find the fastest* encryption method. Rather, we argue that Boneh and Franklin’s scheme on which our SIBE scheme is based provides fairly efficient, sound in terms of security [32], and relatively easily implementable encryption scheme for ADS-B system. We leave improving the efficiency of SIBE scheme while providing the same level of security as future research.

V. CONCLUSION

In the new e-Enabled aircraft, advanced network technologies play a vital role in enhancing flight safety and efficiency. The ADS-B system is one of the essential components of the e-Enabled aircraft, whether they are commercial or military ones.

Security for ADS-B is of great importance in that in the e-Enabled aircraft communications, the airspace is now considered as a cyberspace where aircraft act as intelligent nodes, vulnerable to various cyber attacks. In this paper, we provided confidentiality framework for ADS-B which is based on the proposed SIBE (Staged Identity-Based Encryption) scheme. The security analysis and the performance analysis based on implementation for the SIBE scheme show that our confidentiality framework is sound and meets the security and efficiency requirements. Our future research includes integrating the confidentiality framework with our previous framework for integrity and authentication of ADS-B data through identity-based signature [14].

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